[2]:
```python
number = 6
squared = number * number
print(squared)
```

```
36
```

[3]:
```python
a=10
b=20
print("addition:",a+b)
print("subtraction:",a-b)
print("multiplication:",a*b)
print("division:",a/b)
```

```
addition: 30
subtraction: -10
multiplication: 200
division: 0.5
```

[5]:

```python
a = 5
b = 10

print("Before swapping:")
print("a =", a, "b =",b)
temp = a
a = b
b = temp

print("After swapping:")
print("a =", a, "b =", b)
```

```
Before swapping:
a = 5 b = 10
After swapping:
a = 10 b = 5
```

[6]:

```python
# Average of three numbers

a = 10
b = 20
c = 30
```

```python
average = (a + b + c) / 3

print("The average is:", average)
```

```
The average is: 20.0
```

[7]:

```python
celsius = 37

fahrenheit = (celsius * 9/5) + 32

print("Temperature in Fahrenheit:", fahrenheit)
```

```
Temperature in Fahrenheit: 98.6
```

[8]:

```python
num = int(input("Enter a number: "))

if num > 0:
    print("Positive")
elif num < 0:
    print("Negative")
else:
    print("Zero")
```

```python
average = (a + b + c) / 3

print("The average is:", average)
```

```
The average is: 20.0
```

[7]:

```python
celsius = 37

fahrenheit = (celsius * 9/5) + 32

print("Temperature in Fahrenheit:", fahrenheit)
```

```
Temperature in Fahrenheit: 98.6
```

[8]:

```python
num = int(input("Enter a number: "))

if num > 0:
    print("Positive")
elif num < 0:
    print("Negative")
else:
    print("Zero")
```

```
Enter a number:  45
Positive
```

[9]:
```python
year = int(input("Enter a year: "))

if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
    print("Leap Year")
else:
    print("Not a Leap Year")
```

```
Enter a year:  23
Not a Leap Year
```

[10]:
```python
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
op = input("Enter operator (+, -, *, /): ")

if op == '+':
    print("Result =", a + b)
elif op == '-':
    print("Result =", a - b)
elif op == '*':
    print("Result =", a * b)
elif op == '/':
    if b != 0:
        print("Result =", a / b)
```

```
    else:
        print("Division by zero not allowed")
else:
    print("Invalid operator")
```

```
Enter first number:  23
Enter second number:  45
Enter operator (+, -, *, /):  +
Result = 68.0
```

[12]:
```
ch = input("Enter a character: ").lower()

if ch in 'aeiou':
    print("Vowel")
elif ch.isalpha():
    print("Consonant")
else:
    print("Not an alphabet")
```

```
Enter a character:  kushal
Consonant
```

[13]:
```
marks = int(input("Enter marks: "))

if marks >= 90:
    grade = "A"
```

```
elif marks >= 75:
    grade = "B"
elif marks >= 50:
    grade = "C"
elif marks >= 35:
    grade = "D"
else:
    grade = "F"

print("Grade:", grade)
```

```
Enter marks:  90
Grade: A
```

[14]:

```
num = int(input("Enter a number: "))

if num % 5 == 0 and num % 11 == 0:
    print("Divisible by both 5 and 11")
else:
    print("Not divisible by both 5 and 11")
```

```
Enter a number:  69
Not divisible by both 5 and 11
```

[15]:
```python
ch = input("Enter a character: ")

if ch.isalpha():
    print("Alphabet")
elif ch.isdigit():
    print("Digit")
else:
    print("Special Character")
```

```
Enter a character:  chandan
Alphabet
```

[16]:
```python
num = int(input("Enter a number: "))
fact = 1

for i in range(1, num + 1):
    fact *= i

print("Factorial =", fact)
```

```
Enter a number:  56
Factorial = 71099858780486345185404564746372494973649797888116845868744704000000000000
```

[17]:
```python
num = int(input("Enter a number: "))
```

```python
if num % 3 == 0:
    print("Divisible by 3")
else:
    print("Not divisible by 3")
```

```
Enter a number:  67
Not divisible by 3
```

[18]:
```python
p = float(input("Enter principal: "))
r = float(input("Enter rate of interest: "))
t = float(input("Enter time (years): "))

si = (p * r * t) / 100

print("Simple Interest =", si)
```

```
Enter principal:  34
Enter rate of interest:  67
Enter time (years):  90
Simple Interest = 2050.2
```